

DEMESHWAR RANA

Data Science Student | Machine Learning & Deep Learning | Data Visualisation

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Work Authorisation: Student Residence Permit (Germany) — eligible for Werkstudent roles (up to 20 hrs/week)

PROFILE

MSc Data Science & AI student at Gisma University (Potsdam) specialising in machine learning and deep neural networks (CNN, RNN/GRU/LSTM, transfer learning). Hands-on experience training and rigorously validating models with TensorFlow/Keras and Scikit-learn, from raw timeseries and image data through to a result that holds up under proper testing. Seeking a Werkstudent role in Data Science, Machine Learning, or AI Engineering.

TECHNICAL SKILLS

Machine Learning & Deep Learning: Neural Networks (CNN, RNN, GRU, LSTM, GANs, Transformers/Attention), Transfer Learning, Regularisation (Dropout, BatchNorm, EarlyStopping, Data Augmentation), Scikit-learn, XGBoost, Random Forest, Regression Analysis

Frameworks: TensorFlow, Keras | **Languages:** Python | **Data Processing:** Pandas, NumPy

Data Visualisation: Plotly, Matplotlib, Seaborn | **Tools & Deployment:** Streamlit, GitHub, Jupyter Notebook

Engineering (prior background): SolidWorks, Siemens NX (Unigraphics) — 3D Modelling, Motion & Stress Analysis

EDUCATION

M.Sc. in AI, Data Science & Digital Business | Gisma University of Applied Science, Potsdam, Germany Jan 2026 – Jan 2028 (expected)

- Key modules: Machine Learning, Neural Networks, Business Analytics, Data-Driven Decision Making, Digital Business Strategy

B.Tech in Mechanical Engineering | Jawaharlal Nehru Government Engineering College, Sundernagar, India Aug 2021 – Jun 2024

- Strong grounding in analytical thinking, systems design, and quantitative problem-solving

Diploma in Mechanical Engineering | Government Polytechnic College, Sundernagar, India 2018 – 2021

WORK EXPERIENCE

Robotics & Data Intern | Centre of AI and Robotics – IIT Mandi, India Jan 2024 – May 2024

- Identified and resolved a structural weak point in a quadruped robot frame through motion and stress analysis in SolidWorks — catching the issue before fabrication and avoiding a costly re-design
- Engineered a complete cleaning assembly for a ship hull robot using mechanism inversion to achieve precise 45° brush oscillations, enabling reliable automated hull cleaning without manual rework

MACHINE LEARNING & DEEP LEARNING PROJECTS

Day-Ahead Electricity Demand Forecasting (National Grid ESO, GB)

Python · TensorFlow/Keras (SimpleRNN, GRU, LSTM) · Pandas · Scikit-learn · Plotly

- Framed as a grid-operations problem: given 24 hours (48 half-hourly readings) of demand, generation-mix, and calendar data, predict national demand 48 half-hour steps (one full day) ahead to support GB's day-ahead generation scheduling
- Sourced 16 years of half-hourly ESO data (2009–2025) but trained on the post-2022 window only, avoiding a temporal distribution shift from the full historical span
- Benchmarked SimpleRNN, GRU, and LSTM architectures under identical conditions (fixed seed, Dropout, EarlyStopping, time-ordered 80/10/10 split); selected GRU as the final model
- Achieved a test R^2 of 0.87 (MAE 1,708 MW), consistently outperforming a naive persistence baseline (R^2 0.79)

Image Classification: CNN & Transfer Learning (EfficientNetB0)

Python · TensorFlow/Keras · CNN · EfficientNetB0 · Matplotlib

- Built a regularised CNN classification pipeline on CIFAR-10 using Dropout, BatchNorm, and data augmentation to control overfitting
- Applied transfer learning with a pretrained EfficientNetB0 (ImageNet weights): frozen-base head training followed by fine-tuning of top layers at a low learning rate
- Improved test accuracy meaningfully over the from-scratch CNN baseline, while also reducing total training time

Code: github.com/DemeshwarRana/image-classification-pipeline/blob/main/efficientnet_transfer_learning.ipynb

LANGUAGES & ADDITIONAL INFORMATION

Languages: Hindi (Native) | English (B2) | German (A2 — actively improving)

Continuous Practice:

- Builds and iterates on end-to-end ML/DL pipelines (data ingestion → preprocessing → model training → evaluation) on personal projects on a near-daily basis to stay hands-on with new architectures and tools

Activities:

- Peer mentor in Data Visualisation — helped fellow students apply Plotly, Matplotlib, and Seaborn to their own coursework datasets
- Prior experience as Sales Executive — developed client-facing communication and negotiation skills through direct customer engagement